

9/20

* condition

1. Write a program to check whether a number is positive, negative, or zero.

```
num = int(input("enter the number"))
```

```
if num>0:
    print("number is positive")
elif num<0:
    print("number is negative")
else:
    print("zero")
```

enter the number 0

zero

1. Write a program to find the largest of three numbers.

```
num1 = int(input("enter the first number"))
```

```
num2 = int(input("enter the second number"))
```

```
num3 = int(input("enter the third number"))
```

```
if num1>num2 and num1>num3:
    print(num1,"is larger than others")
elif num2>num1 and num2>num3:
    print(num2,"is larger than others")
elif num3>num1 and num3>num2:
    print(num3,"is larger than others")
else:
    print("invalid input")
```

enter the first number 25
enter the second number 60
enter the third number 20

60 is larger than others

1. Write a program to check if a character is a vowel or consonant.

```
ch = input("enter the character ")
```

```
if ch.lower() in ('a', 'e', 'i', 'o', 'u'):
    print(ch,"the character is vowel ")
elif ch.lower() in ('b', 'c', 'd', 'f', 'g', 'h', 'j', 'k', 'l', 'm', 'n', 'p', 'q', 'r', 's', 't', 'v', 'x', 'y', 'z'):
    print(ch,"the character is consonant ")
else:
    print("ivaild input")
```

enter the character i

i the character is vowel

1. Write a program to calculate the electricity bill:

- First 100 units → ₹5/unit
- Next 100 units → ₹7/unit
- Above 200 units → ₹10/unit

```
unit = int(input("enter the consumed unit"))
```

```
if unit<=100:
    bill= unit*5
    print("total bill",bill)
elif unit>100 and unit<=200:
    bill= unit*7
    print("total bill",bill)
elif unit>200:
    bill= unit*10
    print("total bill",bill)
else:
    print("invalid input")
```



enter the consumed unit 500

total bill 5000

* Loops (for / while)

1. Print all even numbers from 1 to 100.

```
for i in range(1,101):
    if i % 2 == 0:
        print(i)
```



2
4
6
8
10
12
14
16
18
20
22
24
26
28
30
32
34
36
38
40
42
44

46
48
50
52
54
56
58
60
62
64
66
68
70
72
74
76
78
80
82
84
86
88
90
92
94
96
98
100

1. Find the factorial of a given number using a loop.

```
n=int(input("enter the number"))  
fact=1  
for i in range(1,n+1):  
    fact=fact*i  
print(f"factorial of {n} = {fact}")
```

enter the number 7

factorial of 7 = 5040

1. Find the sum of digits of a given number.

```
n =int(input("enter the number"))  
sum=0  
for i in range(1,n+1):  
    sum=sum+i  
print(f"sum upto {n} = {sum}")
```

enter the number 11

sum upto 11 = 66

* Functions

1. Write a function to find the greatest of three numbers.

Recursion,
DSA,

```
def greatest(num1,num2,num3):
```

```
    if num1>num2 and num>num3:
        print(num1,"is larger than others")
    elif num2>num1 and num2>num3:
        print(num2,"is larger than others")
    elif num3>num1 and num3>num2:
        print(num3,"is larger than others")
    else:
        print("invalid input")
```

```
num1 = int(input("enter the first number"))
```

```
num2 = int(input("enter the second number"))
```

```
num3 = int(input("enter the third number"))
```

```
greatest(num1,num2,num3)
```

```
enter the first number 46
```

```
enter the second number 78
```

```
enter the third number 98
```

```
98 is larger than others
```

1. Write a function to check whether a number is prime.

```
def prime(n):
```

```
Cell In[45], line 2
```

```
^
```

```
_IncompleteInputError: incomplete input
```

2. Write a function that returns the reverse of a string.

```
def reverse_string(s):
    return s[::-1]
str1=input("enter a string")
reverse_string(str1)
```

```
enter a string 'swadhin'
```

```
"'nihdaws'"
```